Experimental and Computational Heat Transfer Group

# WHAT ARE THE HEAT FLUX LIMITS OF AIR COOLING?


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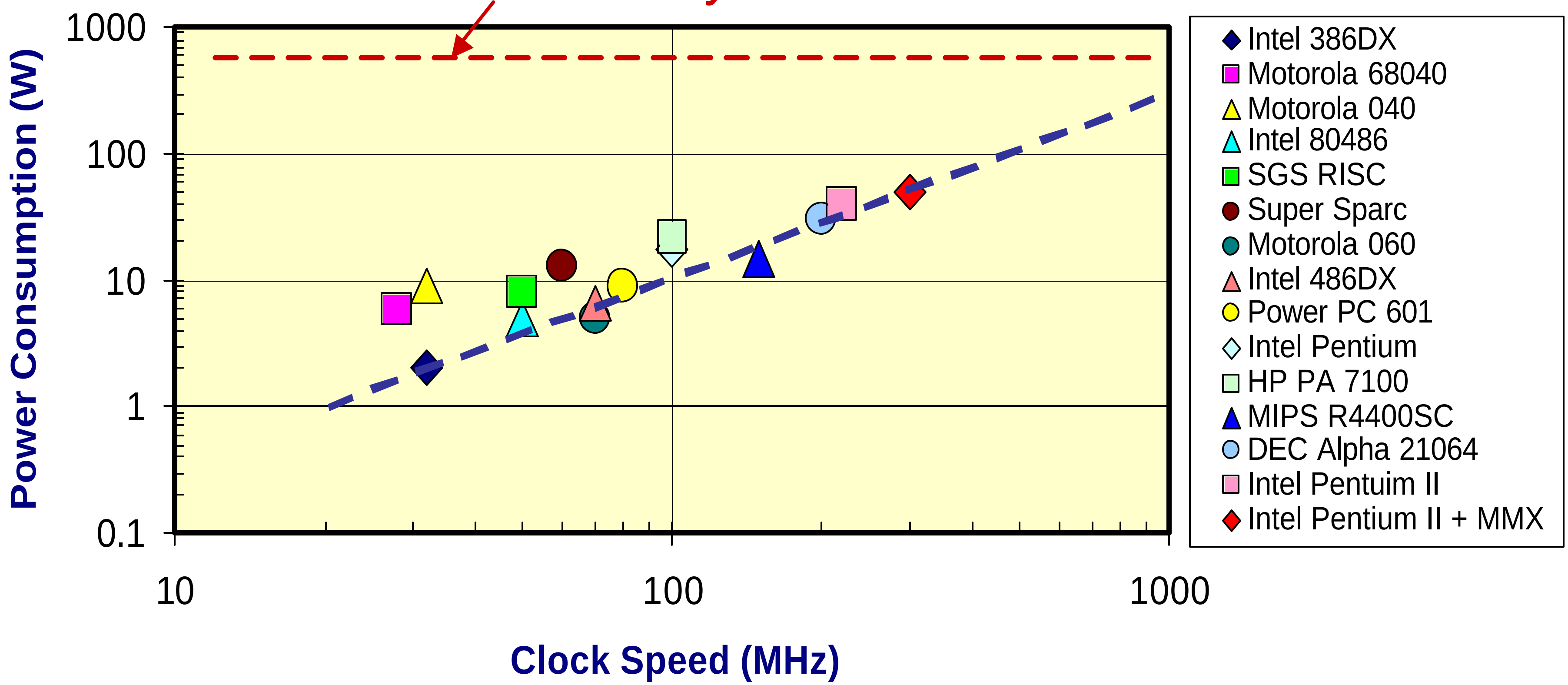
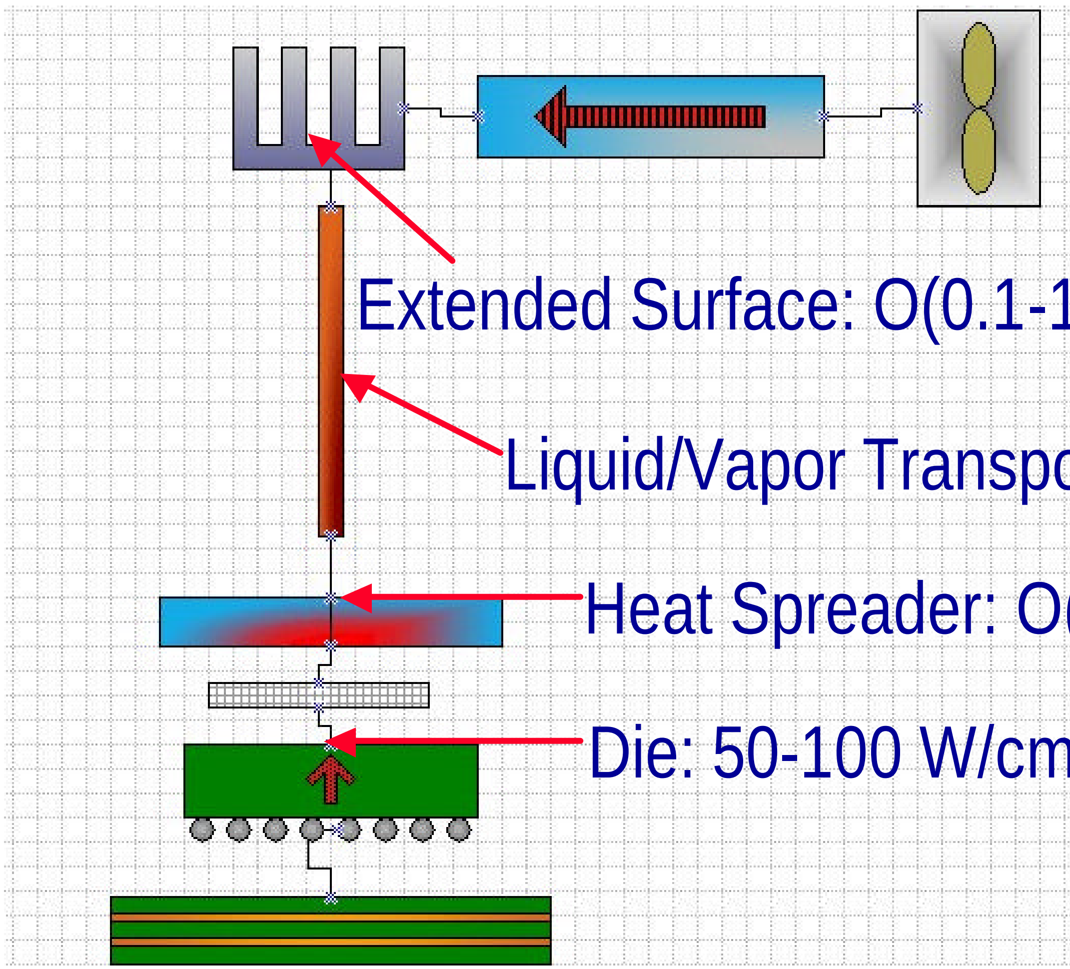
# Thermal Dissipation Trends for Processors

500 W within five years?




After S. Lee, Intel (1998) by permission

# Levels of Heat Flux: TODAY



Heat Sink to Die

Area Ratio: O(1000)

Extended Surface: O(0.1-1) W/cm$^2$ (25 - 250 W/m$^2$-K)

Liquid/Vapor Transporter

Heat Spreader: O(10) W/cm$^2$ (2,500 W/m$^2$-K)

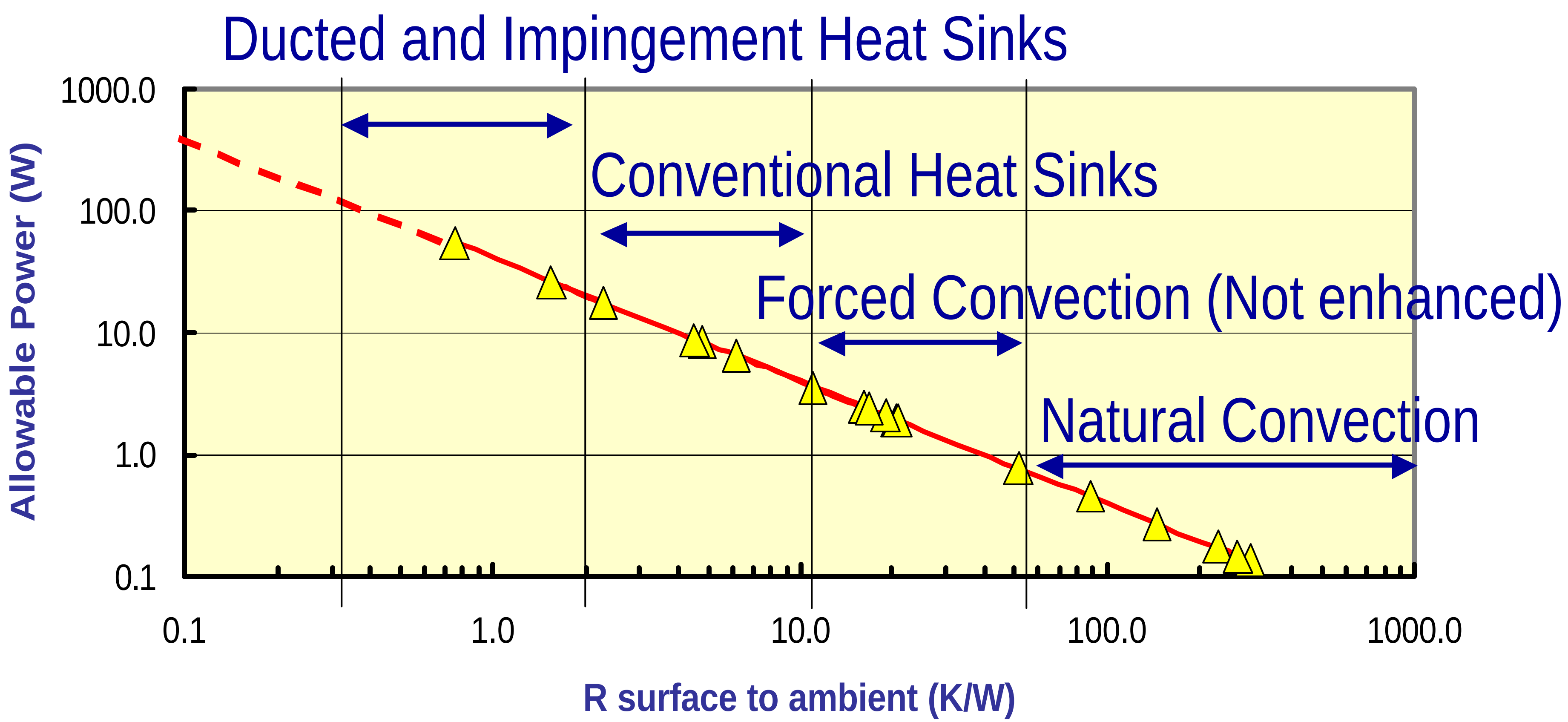
Die: 50-100 W/cm$^2$ (25,000 W/m$^2$-K)

# Achievable Power Consumption for Various Cooling Modes

| Study | Year | Geometry | Velocity (m/s) | had (W/m2-K) | Aplanform (cm2) | R (K/W) | Q (W) |
|---|---|---|---|---|---|---|---|
| Ortega and Lall | 2001 | 1.0 inch heat source in horizontal encl on FR4 | natural convection | 7 | 6.3 | 228.6 | 0.2 |
| Horton | 1981 | Grooved Plates (dips) | natural convection | 5.5 | 6.3 | 290.9 | 0.1 |
| Horton | 1981 | Grooved Plates (flat packs) | natural convection | 6 | 6.3 | 266.7 | 0.2 |
| Ortega and Moffat | 1985 | 0.5 inch cubes (no opposing plate) | natural convection | 8.5 | 8.1 | 145.9 | 0.3 |
| Ortega and Moffat | 1985 | 0.5 inch cubes (first row H/B=1.5) | bifc | 14 | 8.1 | 88.6 | 0.5 |
| Ortega and Wirth | 1993 | Strip heat source (laminar flow) | low velocity | 31 | 6.3 | 51.6 | 0.8 |
| Ortega and Wirth | 1993 | Strip heat source (turbulent flow) | high velocity | 78 | 6.3 | 20.5 | 2.0 |
| Lehmann and Wirtz | 1984 | Ribs in crossflow | ReH1=12000 | 36 | 25.0 | 11.1 | 3.6 |
| Wirtz and Dykshoorn | 1984 | Sparse Flatpacks | Rechannel=3700 | 48 | 12.9 | 16.1 | 2.5 |
| Arvizu and Moffat | 1981 | Sparse Cubes | 1.5 m/s | 60 | 8.1 | 20.7 | 1.9 |
| Arvizu and Moffat | 1981 | Dense Cubes | 1.5 m/s | 65 | 8.1 | 19.1 | 2.1 |
| Arivzu and Moffat | 1981 | Dense Cubes | 5 m/s | 200 | 8.1 | 6.2 | 6.5 |
| Ortega and Singer | 1997 | Impinging Slot Jet | low velocity | 95 | 6.3 | 16.8 | 2.4 |
| Ortega and Singer | 1997 | Impinging Slot Jet | high velocity | 330 | 6.3 | 4.8 | 8.3 |
| Ortega and Urdaneta | 2001 | Pin Fin Heat Sink - Side In Side Out | 1.5 m/s | | 6.3 | 4.5 | 8.9 |
| Ortega and Urdaneta | 2001 | Pin Fin Heat Sink - Side In Side Out | 6.6 | | 6.3 | 1.6 | 25.6 |
| Ortega and Issa | 2001 | Pin Fin Heat Sink - Top In Side Out | 1.94 | | 6.3 | 2.3 | 17.4 |
| Ortega and Issa | 2002 | Pin Fin Heat Sink - Top In Side Out | 20 | | 6.3 | 0.8 | 53.3 |

# Achievable Power Consumption for Various Cooling Modes

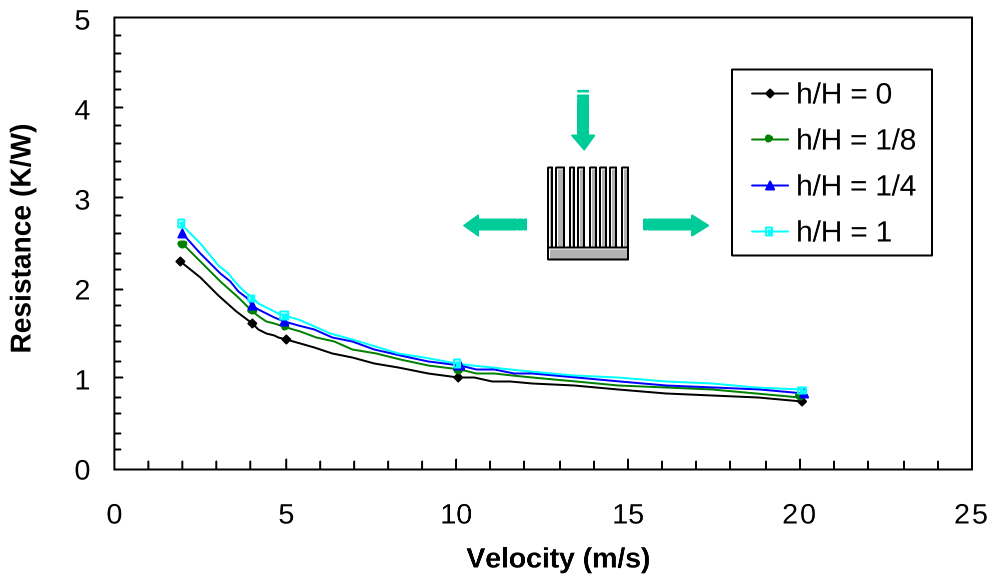
Nominal Ts-Ta = 40 C  Planform area = 2.5 x 2.5 cm$^2$



# Observations

- Air cooling limits have not been challenged based on physics, but by FORM FACTOR
  - It is <u>not difficult</u> to achieve h = O(25-250)
  - It <u>may be difficult</u> to achieve a heat sink to die area ratio of O(1000)
    - The first level of heat spreading is increasingly difficult (vapor chambers)
    - Overall heat sink volume is highly constrained in portable and desktop systems

# Physical Limits




- Aggressive Jet Impingement over large area Pin Fin Heat Sinks will achieve R =O(0.1 to 0.5) K/W
- Significant pressure head O(1 to 2 in. water)
- Diminishing returns as impingement velocity increases
- Acoustic noise